# EVERY ETALE COVERING IS AN FPQC COVERING

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The proof is the author's complete argument for both defining properties, with the finite refinement written out.

*Editorial note.* The label lemma-etale-fpqc occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card.

*Editorial note.* The single labelled reference is the author's parenthetical pointer to a chapter of the Project not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

*Editorial note.* The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

*Allowed standard prerequisites (editorial).*

- Etale morphism of schemes and its flatness and openness. Same chapter, the properties of etale morphisms used throughout: they are flat and open. Supplies both the faithful flatness of the members of the covering and the openness used to write the target as a union of images.
- fpqc covering: faithfully flat and quasi-compact. Same chapter, the definition of an fpqc covering as a family of flat morphisms whose images cover the target and which is quasi-compact as a family. Supplies the two properties verified in the proof.
- Quasi-compactness of an affine open. Same chapter, the standard fact that an affine scheme is quasi-compact. Supplies the finite refinement of the open covering of an affine open V.

*Source reference.* Original lemma 022C: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 2741–2741. Cross-chapter navigation pointer occurring in the author's parenthetical 'see also' at proof lines 2739-2741, which points the reader to the corresponding statement in the chapter Topologies. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

**Lemma 0.1.** *Any étale covering is an fpqc covering.*

*Proof.* (See also Topologies, Lemma 022C.) Let $\{\varphi_i : U_i \to U\}_{i \in I}$ be an étale covering. Since an étale morphism is flat, and the elements of the covering should cover its target, the property fp (faithfully flat) is satisfied. To check the property qc (quasi-compact), let $V \subset U$ be an affine open, and write $\varphi_i^{-1}(V) = \bigcup_{j \in J_i} V_{ij}$

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for some affine opens $V_{ij} \subset U_i$. Since $\varphi_i$ is open (as étale morphisms are open), we see that $V = \bigcup_{i \in I} \bigcup_{j \in J_i} \varphi_i(V_{ij})$ is an open covering of $V$. Further, since $V$ is quasi-compact, this covering has a finite refinement. $\square$